

End-of-Life Plan

This prototype was designed as a modular system. The main subsystems can be separated from each other, including the mixing chamber, the syringe pumping system, the pH sensing system, the CoreXY deposition gantry, the heating system, the UV crosslinking system and the control electronics. Because of this modular design, the preferred end-of-life route is not disposal. The priority is to keep the prototype, or its useful modules, for future student projects and further research.

The first option is to hand over the complete prototype to future students. This is the preferred route if the device is still structurally safe and the main modules are still working. Future students can continue the development of the full hydrogel fabrication process, including real collagen testing, improved pH calibration, gantry accuracy validation, UV safety improvement and closed-loop process control. The submitted CAD files, code, risk assessments and unused material list should be kept with the prototype so that the next users can understand the current design status.

The second option is to let the university reuse separate modules for other teaching or research projects. If the complete prototype is not needed, the system should be dismantled into reusable modules. For example, the CoreXY gantry can be reused for motion control or deposition projects. The Arduino boards, sensors and camera modules can be reused for control, image processing or sensing projects. Motors, bearings, screws and brackets can also be reused in future mechanical prototypes.

If a component cannot be reused, it should be recycled or disposed of through the correct route. Electrical parts should not be placed in normal waste. Parts that have contacted collagen, NaOH, riboflavin or buffer solution should only be reused if they can be cleaned safely. If they are contaminated, damaged or difficult to clean, they should be disposed of through the appropriate laboratory waste route.

Position: MechSpace, Plastic Box B42 and B41. Collagen Solution in 3rd floor.

Prototype part or module	Preferred end-of-life action	Alternative action if not reused
Complete prototype	Keep as a full system for future students to continue the project. It can support further work on collagen mixing, pH control, gel deposition, UV crosslinking and validation.	If the full system is no longer needed, separate it into reusable modules before any disposal is considered.
Mixing chamber	Keep for future hydrogel or temperature-sensitive fluid projects. The chamber can be reused, syringe- based preparation or mixing tests.	Return to the department or dispose of through the approved electrical appliance route if it is damaged.
Syringe pumps, valve system and fluidic routing	Reuse for future microfluidic, mixing or dosing projects. Clean parts should be kept with the prototype as spare fluidic modules.	Tubing, syringes and connectors that are used, worn or contaminated should be disposed of through the correct lab waste route.
CoreXY deposition gantry	Reuse as a motion platform for deposition, printing, scanning or positioning projects. Motors, belts, rails, bearings and	If dismantled, working motors and mechanical parts should be stored for future projects. Damaged plastic or acrylic parts

Prototype part or module	Preferred end-of-life action	Alternative action if not reused
	brackets should be kept if functional.	should be disposed of according to workshop procedure.
pH sensing and optical sensing system	Reuse the camera, colour sensor, LEDs and control board for future pH calibration, colour detection or image processing work.	If electronic parts fail, dispose of them through the electronic waste route. Optical mounts or brackets can be reused if still intact.
Control electronics and wiring	Reuse Raspberry Pi, Arduino boards, ESP32 boards, motor drivers, breadboards and sensors in future student projects. Labelled wiring should be kept if it is still safe and clear.	Damaged boards, loose unsafe wires or broken electrical parts should be removed and disposed of as electronic waste.
Heating system	Reuse the heating mat and temperature control parts for future gelation or temperature-control experiments if they pass safety checks.	If the heating mat, relay or power connection is damaged, it should not be reused. It should be disposed of through the correct electrical waste route.
UV crosslinking system	Keep the UV lamp, enclosure and shielding for future controlled crosslinking tests if the system is still safe. The enclosure should remain labelled as a UV safety component.	If the UV lamp or shielding is damaged, the system should not be operated. The lamp and electrical parts should be disposed of through the correct route.
Structural frame, acrylic panels and 3D printed parts	Reuse intact panels, brackets and printed parts for future prototype changes. The modular design allows parts to be removed and replaced without rebuilding the full system.	Damaged acrylic, PLA or other plastic parts should be disposed of through the workshop waste route. Metal parts should be recycled where possible.
Fluid-contact parts exposed to collagen, NaOH, riboflavin or buffer	Reuse only if the parts can be cleaned safely and are still suitable for laboratory use. Cleanable parts should be labelled before storage.	Contaminated, blocked or single-use parts should be disposed of through the approved lab waste route. They should not be returned to general storage.
Documentation and digital files	Keep CAD files, code, risk assessments, test results, calibration data and operating notes with the physical prototype. This helps future users understand what has been built and what still needs validation.	If the prototype is dismantled, the files should still be stored by the department or project supervisor as a record of the design and testing work.

Clean unused materials are listed separately in the unused material list. These materials should be kept for future student projects where possible. If any unused material becomes damaged, expired or unsuitable for reuse, it should be disposed of through the relevant workshop, laboratory or electronic waste route.

Overall, the recommended end-of-life approach is to reuse first, then recycle, and only dispose of parts when reuse or recycling is not suitable. This approach reduces waste and also allows future students or the school to gain value from the modules, components and documentation developed in this project.